

R Notebook

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats   1.0.0      v readr     2.1.5
## v ggplot2    4.0.0      v stringr  1.5.1
## v lubridate  1.9.4      v tibble   3.3.0
## v purrr      1.0.4      v tidyr    1.3.1

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##   group_rows
```

```
cat("\\captionsetup[labelformat=empty]")
```

```
primer_data <- read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_all/prim
primer_tbl = primer_data %>%
  mutate(
```

```

primer_length = 15,
num_prim_kb = round(num_prim_kb, 2),
num_ref_kb = round(num_ref_kb, 2),
virus = recode(virus,
  `pastv` = "PAstV",
  `ev-g` = "EV-G",
  `hpv18` = "HPV18",
  `pbov` = "PBoV",
  `pcv` = "PCV",
  `psapev` = "PSapeV",
  `ptev` = "PTeV",
  `pkv` = "PKV",
  `psapov` = "PSapoV",
)
) %>%
arrange(virus) %>%
select(virus, primer_length, num_primers, num_fwd, num_rev, num_refs, num_prim_kb, num_ref_kb) %>%
rename(
  "Primer length (nt)" = primer_length,
  "Virus" = virus,
  "Forward" = num_fwd,
  "Reverse" = num_rev,
  "References" = num_refs,
  "Primers / kb genome" = num_prim_kb,
  "Refs / kb genome" = num_ref_kb,
  "Total Primers" = num_primers,
) %>%
kable(
  format = "latex",
  booktabs = TRUE,
  caption = "List of primers and targets designed for MSSPE-based enrichment",
  digits = 3,
  align = "lrrr", # adjust as needed: text left, numbers right
  format.args = list(big.mark = ",")
) %>%
kable_styling(
  latex_options = c("striped", "hold_position"), # polished journal look
  position = "center",
  full_width = TRUE,
  font_size = 9
) %>%
row_spec(0, bold = TRUE) %>% # strong header
collapse_rows(columns = 1, latex_hline = "major") # if repeating target names

primer_tbl

```

List of primers and targets designed for MSSPE-based enrichment

Virus	Primer length (nt)	Total Primers	Forward	Reverse	References	Primers / kb genome	Refs / kb genome
ASFV	15	212	106	106	122	1.18	0.68
CSFV	15	109	47	62	219	8.86	17.80
EV-G	15	116	56	60	44	15.77	5.98
FMDV	15	48	22	26	72	5.78	8.67
HPV18	15	5	1	4	2	6.29	2.52
IAV	15	631	292	339	3,613	46.74	267.63
NiV	15	111	61	50	80	6.17	4.44
PAstV	15	52	21	31	16	7.99	2.46
PBoV	15	37	20	17	16	7.12	3.08
PCV	15	17	10	7	191	9.62	108.09
PKV	15	61	24	37	64	7.53	7.90
PRRSV	15	64	32	32	118	4.18	7.71
PSapeV	15	116	53	63	40	15.43	5.32
PSapoV	15	50	32	18	21	6.70	2.82
PTeV	15	128	52	76	27	17.76	3.75